VADIM FEDOROV

Electrical Engineering Student | Embedded Robotics and Neurotechnology
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EDUCATION

University of California, Davis Expected June 2028
Bachelor of Science in Electrical Engineering | GPA 3.7 | Dean's List Fall 2025 and Winter 2025

RESEARCH EXPERIENCE

Laboratory for AI, Robotics, and Automation (LARA) Jan 2026 - Present
Undergraduate Robotics Researcher | University of California, Davis

- Restored continuous control of a 19-actuator robotic hand after roughly two-second teleoperation failures by diagnosing I2C bus contention and rewriting inbound and outbound microcontroller firmware; achieved approximately 50 ms end-to-end latency.
- Designed and tested custom control PCBs from schematic and layout through functional bring-up, adding I2C pull-ups and dedicated voltage regulation; verified simultaneous motor operation at full PWM without freezes or noticeable motion lag.
- Integrated two base-mounted rotary actuators into the PyBullet and micro-ROS pipeline, extending glove-driven teleoperation from hand joints to wrist articulation.

SELECTED PROJECTS

BCI Prosthetic Leg Control Dec 2025 - Jun 2026
Neurotech Club Projects Division | OpenBCI Cyton to Arduino Mega test rig

- Built a Python live EEG decoder using OpenBCI Cyton acquisition through BrainFlow, 8-30 Hz filtering, CSP feature extraction, and LDA scoring; connected decisions to an Arduino Mega serial trigger for bench testing.
- Added dry-run-first output, decision thresholds, refractory timing, and REST-trial blocking; used replay and file-holdout checks to assess generalization before physical actuation.

PyBCI Live-Streaming Interface Jan 2026 - Present
Open-source contribution to EEG research software

- Built PyBCI's live EEG acquisition and streaming backend, connecting board data to the real-time research interface.
- Implemented the frontend's stacked live EEG plots and start, pause, save, and stop recording controls.

TECHNICAL SKILLS

Programming: C/C++, embedded C, Python, Java, MATLAB
Electronics: KiCad, PCB schematic and layout, board bring-up, I2C, Teensy, sensor integration, soldering
Robotics and research: micro-ROS, PyBullet, BrainFlow, OpenBCI Cyton, EEG signal processing, Git, Linux

ADDITIONAL EXPERIENCE

Cherry Chase Pet Hospital Jun 2024 - Sep 2025
Customer Service Representative | Sunnyvale, CA

- Managed client communication across phone, email, and front desk; coordinated appointments and intake with veterinary staff during peak clinic hours.

PUBLICATION

Published The Implementation of Ethical AI Within Autonomous Vehicles in the Journal of Student Research, 2024.